# supplementary materials - Reconciling latent variables and networks: Exploring and extending the Psychometric-Toolbox

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## Extending the Psychometric-Toolbox

### PGM, KIM, VDAR, GRG, (T)ERGM

Let's start by considering PGMs[1] again and assume that variable interactions result in the coordination of variable states, six such base models for different variable types are presented in Finnemann et al. (2026) and include the IM for binary variables. The IM has also been extended in econometrics into the time-series domain with *Multiple Time Series IM* (Takaishi, 2015, 2016) and in physics with the *Kinetic IM* (KIM; Aguilera et al., 2021) which has been formally related to the *Vector Discrete AR* (VDAR; Campajola et al., 2021) model, the discrete analogue to the well-known VAR model[2]. Furthermore, the ER model has been generalized with *Generalized Random Graph* (GRG[3]; Shi and Qian, 2009) and extended in the *Exponential Random Graph Model* (ERGM[4]; Q. Zhang and Liang, 2024), which has with the *Temporal ERGM* (TERGM; Hanneke et al., 2010) a time-series extension as well and can also be related back to the KIM (Campajola et al., 2022).

### PGMs — Random Graph Models (Graph Theory)

The Idiographic IM presents an interesting formal relation between PGMs and Random Graph models from Graph Theory usually applied in **Network Science** (Coscia, 2025; Newman, 2003). These are research fields that have to a large extent developed independent from each other, which makes it sometimes unclear how they relate (Hidalgo, 2016; Iñiguez et al., 2020; Sweet & Wang, 2025). This is an issue illustrated by the common use of node centrality measures for quantifying relative importance of nodes in network psychometrics (Borsboom et al., 2021), which has been criticized because these centrality measures were initially proposed for (unweighted) random graph models (L. F. Bringmann et al., 2019; Hallquist et al., 2021; Spiller et al., 2020). But they could conceivably be applied to weighted networks as well (Opsahl et al., 2010) and due to this relation therefore also to PGMs such as the IM or GGM.

Graph Theory (R. Das & Soylu, 2023; Diestel, 2025) is a branch of mathematics that studies graphs, a mathematical structures for modeling (pairwise) relationships between objects. A graph $G$ is typically defined as an ordered pair $G = (V, E)$ where $V$ is a set of vertices (or nodes/variables), and $E \subseteq \{(i, j) \mid i, j \in V\}$ is a set of edges, which are pairs of vertices representing connections between them. A common distinction we have already

[1] the term 'Probabilistic Network Models (PNMs)' is also used in psychometrics (Finnemann et al., 2026) and other fields (e.g., Climatology (Graafland et al., 2020), Bioinformatics (Graafland & Gutiérrez, 2022)), but since PGMs are an established term in Machine Learning for such models, maybe we should reserve the term PNMs more generally for models with aspects of both Random Graph Models and PGMs, but again, the boundary is fuzzy here as well (Farasat et al., 2015; Marsman & Huth, 2023).

[2] $\Rightarrow$ IM (binary MRF) $\rightarrow$ KIM $\Longleftrightarrow$ VDAR — VAR $\rightarrow$ gVAR $\leftarrow$ GGM (gaussian MRF); Autoregressive and state-space models are GMRFs (see Rue & Held, 2005).

[3] Temporally Generalized Random Graphs (TGRG; Mazzarisi et al., 2020), Discrete Auto Regressive Temporally Generalized Random Graphs (DAR-TGRG; Mazzarisi et al., 2020)

[4] Related to GLMs (Kolaczyk & Csárdi, 2020) and can be re-defined as a PGM (Farasat et al., 2015).

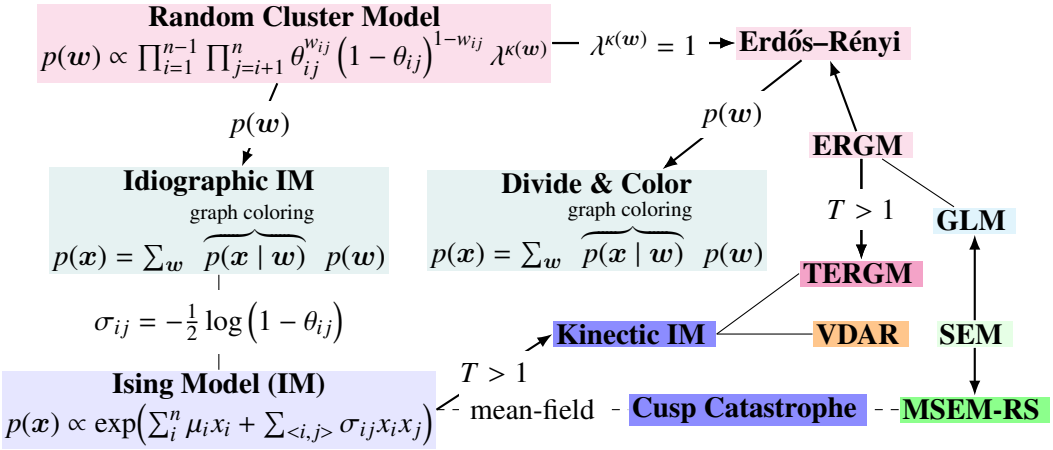



**Figure 8**

*Erdős–Rényi (ER or Random Graph Model), Exponential Random Graph Model (ERGM), Temporal Exponential Random Graph Model (TERGM), Vector Discrete Autoregressive (VDAR), Mixture SEM with Regime-Switching (MSEM-RS). Areas of Research: Psychometrics / , Network Psychometrics , Econometrics , Statistics & Machine Learning , Graph Theory & Network Science / , Physics / (light colored for cross-sectional and darker colored for time-series data)*

encountered previously is between directed graphs where $E$ consists of ordered pairs of vertices, edges have a direction ($(i, j) \neq (j, i) \rightarrow$ e.g., DAG/BN) and undirected graphs where $E$ consists of unordered pairs of vertices, edges have no direction ($(u, v) = (v, u) \rightarrow$ e.g., IM, GGM). Both can either have weighted edges, meaning each edge $e \in E$ is associated with a weight or strength (e.g, partial-correlation coefficient) or unweighted edges (e.g., ER model). And as previously mentioned, investigating the characteristics of graphs representing psychometrics constructs has been independently proposed again more recently in Marsman and Huth (2023). With some specific application being the use of a graph-theoretic similarity measure for comparing weighted adjacency matrices (Ulitzsch et al., 2023), and *Perturbation Graphs* for assessing experimental interventions and possibly discovering causal relations in psychology (Waldorp et al., 2025). Symmetry properties of graph limits and their implications for studying invariant sub-spaces in dynamical systems are investigated in Bick and Sclosa (2025), which could conceivably provide null-models for correlation networks (Masuda, Boyd, et al., 2025). A interesting discussion about learning Graph structure from time-series data and the duality between predictability and reconstructability is given in Murphy et al. (2024)[5]. Furthermore, in Christensen et al. (2023) a statistic called *weighted topological overlap* (wTO) is presented which quantifies the extent to which two variables have similar partial correlations to other variables in a sparse matrix, Ulitzsch et al., 2023 present a graph-theoretic similarity measure for comparing weighted

[5] $\overset{?}{\leftrightarrow}$ 'Weak vs. Strong Complexity' (Hasselman, 2022) & 'nearly decomposable system vs. non-decomposable system' (de Boer et al., 2021; Rathkopf, 2018)

adjacency matrices, and in Sjobeck et al., 2024 the *Pairwise Approximate Spatiotemporal Symmetry* (PASS) algorithm can identify symmetries in pairwise time-series data.

### IM — Cusp — MSEM-RS

Another interesting connection here might be helpful in addressing the issue of interpretability with increasing model complexity (number of parameters) and can potentially help in enabling advances in research on some conceptual issues in psychology (van der Maas, 2024). In Catastrophe Theory[6] (Montaldi, 2021; Stewart & Peregoy, 1983), continuous and gradual changes in one or more control parameters of a complex system can lead to sudden and discontinuous changes in its dynamics. The Cusp (Mabire-Yon et al., 2026) is a specific type of catastrophe model with two control parameters, a bifurcation parameter (which can cause instability) and an asymmetry parameter (stable linear relationship), and has been applied in psychology to explain characteristics of depression (Cramer et al., 2016), visual perception (Ploeger et al., 2002) and attitude change (van der Maas et al., 2003). Cusp dynamics have also been associated with the Ising model in network psychometrics (Cramer et al., 2016; Finnemann et al., 2021; Loossens et al., 2020; van der Maas et al., 2017), and in the case of a fully connected IM (also called the Curie—Weiss model), with its mean-field behavior (van der Maas, 2024). Nevertheless, its practical use has been constrained by difficulties in performing model fitting and model comparison, to address this issue a modeling framework utilizing Mixture Structural Equation Models (MSEM) for cross-sectional data, or MSEM with regime-switching (MSEM-RS) for panel-data has been proposed in Chow et al. (2015).

### Complex Dynamical Systems - Perspective

When trying to model psychological processes like individual developmental pathways the adoption of a complex dynamical systems perspective can be very beneficial (Hasselman, 2023), and has been invoked in psychometrics for re-conceptualizing depression (Cramer et al., 2016), psychopathology in general (Olthof, Hasselman, Oude Maatman, et al., 2023), change-processes in psychotherapy (Hayes & Andrews, 2020), substance use disorder (Witkiewitz & Tucker, 2025), transitions in mood (Waldorp & Kossakowski, 2020), phenomenology in psychopathology (Kyzar et al., 2025), as well as in educational psychology (Hilpert & Marchand, 2018). This also includes utilizing tool like *Causal Loop Diagrams* (CLDs[7]; Barbrook-Johnson and Penn, 2022; Crielaard et al., 2024) and *Perceived Causal Networks* (PECAN; Burger et al., 2024; Scholten et al., 2025; Vogel et al., 2025) which can be combined with VAR models (Burger et al., 2022).

A clarifying discussion of what is meant by "Complex (Dynamical) System" can be found in Estrada (2024) and Mazzocchi (2025). For an elaboration of how this could be

[6] a branch of Bifurcation Theory (Kuznetsov, 2023) and applied in the study of dynamical systems in **Dynamic Systems Theory** (DST; Connell et al., 2017; Olthof, Hasselman, Oude Maatman, et al., 2023)

[7] e.g., of Depression (Wittenborn et al., 2016)

applied in psychology see van der Maas (2024). Patterns in ILD are likely complex, because EMA data is characterized by complex long-range temporal dependencies (Olthof, Hasselman, Wijnants, & Lichtwarck-Aschoff, 2020; Olthof et al., 2024), and psychological self-ratings exhibit complex dynamics as well (Leising et al., 2025; Olthof, Hasselman, Aas, et al., 2023; Olthof, Hasselman, & Lichtwarck-Aschoff, 2020). Different metrics can be used for quantifying the complexity in time-series data, and George (2025) investigated how they relate to dynamics. Furthermore, this can be tied back very interestingly to statistical models with *Coevolving Latent Space Networks with Attractors* (CLSNA; Pan et al., 2025).

### *Early Warning Signals*

A concrete example of how this this line of research could become relevant in clinical psychology are *Early Warning Signals* (EWS; Evers et al., 2024). A particular EWS is *Critical Slowing Down* (CSD) which is an indication of bifurcations (Evers et al., 2024), and has been applied in psychology to assess how this is related to changes in depression symptoms (Curtiss et al., 2023; Fried et al., 2023) and mood disorder during psychotherapy (Olthof, Hasselman, Strunk, et al., 2020). Specifically, in RNs the geometry of the phase space can function as a EWS (Hasselman, 2022). But more research is needed for the appropriate application of EWS in clinical practice (Cui et al., 2025), current modeling approaches are limited, and theoretical foundations of psychological phenomena have to evolve with methods for data analysis in tandem here as well (Helmich et al., 2024). EWS are also researched in physiology (e.g., (Legault et al., 2024)), where networks in the context of complex dynamical systems are investigated as well (Schöll, 2022).

### Stochastic Block Model

It has also been proposed in psychometrics to combine the *Stochastic Block Model* (SBM[8]; Holland et al., 1983; Peixoto, 2019) with the IM to account for clustering of variables (Sekulovski et al., 2026), in other research areas the SBM and IM have been combined as well (Berthet et al., 2019; Ye, 2021; J. Zhang et al., 2023; Zhao et al., 2021).

### ARMA, DLM, ART

Additionally to the already considered VAR[9] and VDAR model, the ARMA model family is actually much bigger (Holan et al., 2010). The Threshold AR (TAR; Tong and Lim, 1980) has been incorporated into psychometrics specifically for modeling hysteresis in psychological processes (HysTAR; de Jong et al., 2023). More generally again, ARMA models can also be formulated as *Dynamic Linear Models* (DLMs; Carvalho and West, 2007; Laine, 2020), and perhaps interesting for ILD applications is the *Autoregressive*

[8] for relations to other models (see B. Kim et al., 2018)

[9] Variants for ILD: ALS VAR (Bulteel et al., 2016), scGVAR (Park, Chow, et al., 2024; Park, Fisher, et al., 2024), multi-VAR framework (Crawford et al., 2026; Fisher et al., 2022, 2024), mlVAR (L. F. Bringmann et al., 2013; Epskamp et al., 2018; Y. Li et al., 2022)

*Tensor*[10] model (ART; Billio et al., 2023) which can include a HMM for modeling structural changes (Billio et al., 2024).

### EGA, EGM, MNM, MAR

In *Exploratory Graph Analysis* (EGA; H. F. Golino and Epskamp, 2017) a clustering algorithm is applied to a GGM, this has also been extended to *Dynamic EGA* (DynEGA; H. Golino et al., 2022) and generalized with the *Exploratory Graph Model* (EGM; Christensen and Golino, 2024). The GGM has also been extended to include moderation effects in the *Moderation Network Model* (MNM; Haslbeck et al., 2021), both the GGM and MNM are special cases of the *k-order Mixed Graphical Model* (MGM; Haslbeck and Waldorp, 2020; Sedgewick et al., 2016; E. Yang et al., 2014). And *Moderated AR* (MAR; Ernst et al., 2024) models exist as well.

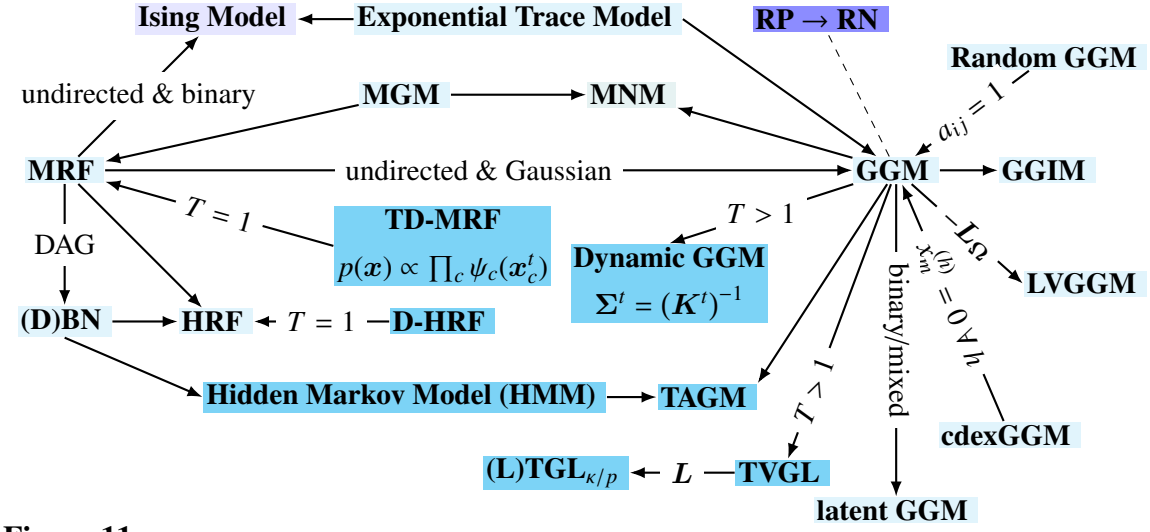



**Figure 11**

*Markov Random Field (MRF), Time-Dynamic MRF (TD-MRF), Dynamic Bayesian Network (DBN), Hybrid Random Field (HRF), Dynamic HRF (D-HRF), Mixed Graphical Model (MGM), Moderated Network Model (MNM), Gaussian Graphical Model (GGM), Gaussian Graphical Interaction Model (GGIM), Latent Variable GGM (LVGGM), ovariate-dependent GGM (cdexGGM), Time Adaptive Gaussian Model (TAGM), Time-Varying Graphical Lasso (TVGL), Latent Variable Time-Varying Graphical Lasso (LTGL). Colors: Network Psychometrics, Statistics & Machine Learning, Physics*

### GGMs in ML (Figure 11)

In the Machine Learning literature many more variants of the GGM have been proposed (*Latent Variable GGM* (LVGGM; R. Li et al., 2023; K. Wang et al., 2023),

[10] Introduction to Tensors (Auddy et al., 2025; Bi et al., 2021). The IM can also be extended to allow for modeling higher-order dependencies with the *p-Tensor IM* (Mukherjee et al., 2022)

*latent GGM* (Zheng et al., 2022), *covariate-dependent GGM* (cdexGGM; J. Wang and Gao, 2025), *Random GGM* (Cai, 2017), *Gaussian Graphical Interaction Model* (GGIM; Fitch, 2019)), with time-series extensions existing as well (*Time-Varying Graphical Lasso* (TVGL; Hallac et al., 2017; Masuda and Lambiotte, 2016), *Group Fused Graphical Lasso* (GFGL; Gibberd and Nelson, 2017), *Latent variable Time-varying Graphical Lasso* (LTGL; Tomasi et al., 2018), *Time Adaptive GGM* (TAGM; Tozzo et al., 2021), *Latent variable Time-varying Graphical Lasso with temporal kernel* ($\text{LTGL}_\kappa$; Tomasi et al., 2021), *Latent variable Time-varying Graphical Lasso with Pattern inference* ($\text{LTGL}_p$; Tomasi et al., 2021)). It seems worthwhile to assess in future work if these models can be applied to psychometric datasets, but this could encounter some challenges (Lavelle-Hill et al., 2025).

**GGM — "causal heuristics"**

When applying GGM's for causal discovery two heuristics pointed out in Ryan et al. (2022) might be useful. The first is that an edge between two variables in the undirected PMRF ($X_i - X_j$) indicates that two variables either share a direct causal link $\left(X_i \rightarrow X_j \text{ or } X_i \leftarrow X_j\right)$ or a common effect $\left(X_i \rightarrow X_k \leftarrow X_j\right)$. Furthermore, the absence of an edge between two variables indicates that these two variables do not share a direct causal link $\left(X_i \nrightarrow X_j \text{ and } X_i \nleftarrow X_j\right)$. Formally it can be stated that a undirected PMRF identifies the moral-equivalence set of the underlying DAG (i.e., collection of all DAG structures that produce the same moral graph, and hence the same PMRF), many different DAGs share the same moral graph (Ryan et al., 2022). GGMs therefore are a natural middle ground between zero-order correlations and DAGs and function as a bridge between the correlational and causal worlds (Epskamp et al., 2017).

**Random Fields (Statistics)**

The GGM and IM have been earlier already connected to the general concept of a Random Field (Hernández-Lemus, 2021), which is basically a joint probability distribution of a set of random variables, of which many variants with different degree of generality exist (see Figure 12). It seems interesting to investigate how Random Fields as the most basic statistical concept, relate to attempts of formalizing a general statistical model taxonomy in (Bürkner et al., 2023) and (Getoor & Taskar, 2007, Chapter 12). Also insightful to note here might be the literature on structure learning in Markov Random Fields (Drton & Maathuis, 2017; Mukherjee et al., 2022; Vogels et al., 2024).

**Beyond Pairwise Correlations - Similarity & Distance Measures**

Outside of psychometrics a large literature on a wide variety of statistical metrics is available as well. Similarity and Distance measures for Probability Density Functions, Vectors, and Graphs (Cha, 2007; S. Das et al., 2021; Josse & Holmes, 2016; Levy et al., 2025; Subramanian et al., 2022; Wills & Meyer, 2020), also more specifically for categorical variables (de van Velden et al., 2024), and temporal graphs (Xiong et al., 2026), as well as for learning relations in time series data more generally (Fulcher & Jones, 2017; Fulcher et

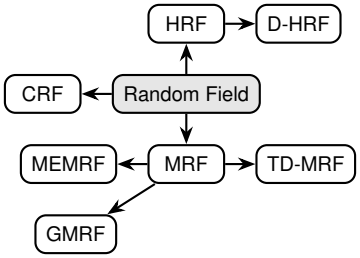



**Figure 12**

*Variants of Random Fields: Markov Random Field (MRF; Hernández-Lemus, 2021), Gaussian MRF (GMRF; Rue and Held, 2005; Sidén and Lindsten, 2020), Mixed Exponential MRF (MEMRF; E. Yang et al., 2014), Time-Dynamic MRF (TD-MRF; Cabello, 2022), Conditional Random Field (CRF; Sutton and McCallum, 2012) Hybrid Random Field (HRF; Kacprzyk et al., 2011), Dynamic Hybrid Random Field (D-HRF; Bongini et al., 2018)*

al., 2013; Lubba et al., 2019), and with different *Pairwise Edge Measure* (PEMs) computed from lagged correlation matrices (Schwarze et al., 2025). In Wulkow (2022) a method for measuring dependencies between variables in a Dynamical System is proposed. Since some of this already intersects, it seems worthwhile to asses in more detail in the future if more of this is relevant and transferable in some form into psychometric applications.

### Dynamic Time Warping

Additionally, some more novel approaches are available as well. *Dynamic Time Warping* (DTW[11]; K. Bringmann et al., 2024; Kopland and Giltay, 2025; Tavenard, 2021) has been developed in speech and signal processing for assessing the alignment between time series data and comes in many forms (Shifaz et al., 2023). Which has recently been applied in psychology as well (de Beurs et al., 2024; Dingemans et al., 2025; Pasteuning et al., 2025; Schwellnus et al., 2025). It can complement highly parametrized approaches (e.g., mlVAR, DSEM) for exploratory identification of interesting patterns when usual assumptions are violated, for example stationarity (Ryan et al., 2025). DTW has been compared to mlVAR with strengths and weaknesses of both approaches assessed in more detail in van der Does et al. (2025), and also combined with PGMs (Y. Li et al., 2018).

### Attractor Landscapes

A useful tool for investigating behavioral change in a complex dynamical system is the attractor[12] landscape (Heino et al., 2023). Along this line of thinking a method for analyzing ILD in the form of vector fields has been proposed with *fitlandr* (Cui, 2025a;

[11] [S1.2.7 Dynamic time warping (dtw)] in (Cliff et al., 2023)

[12] or Energy landscape (Masuda, Islam, et al., 2025), Potential/Stability landscape (Cui, Lichtwarck-Aschoff, et al., 2023; Cui et al., 2026; Hoekstra et al., 2026)

Cui, Hasselman, & Lichtwarck-Aschoff, 2023), and potential landscapes in *simlandr* (Cui, 2025b).

### Recurrence Network

Another based on Recurrence Plots (RPs; http://www.recurrence-plot.tk/) tool is the *Recurrence Network* (RN[13]; Donner et al., 2010, 2011; G. Yang et al., 2019; Zou et al., 2019). RNs provide additional quantitative metrics characterizing the dynamics in time-series data. Obtained by transforming the recurrence matrix of a RP into an adjacency matrix (unweighted and undirected network).

$$A_{t,t'}(\epsilon) = R_{t,t'}(\epsilon) - \delta_{t,t'} \tag{13}$$

where $\delta_{t,t'}$ is the Kronecker delta to avoid artificial self-loops (a point is never connected to itself). Its topological characteristics preserve fundamental phase-space properties of the dynamical system (Donner et al., 2010), and can represent different dynamical regimes (Donges et al., 2015).

Common metrics are the *Local* and *Global Clustering Coefficient* which can provide information about the dimensionality of the underlying process (Varley & Sporns, 2021), the *Average Path Length* characterizes the average phase-space separation of states (Donner et al., 2010), *Transitivity* a measure of dynamical complexity (Zou et al., 2019), and *Assortativity* as a measure of the fragmentation of the attractor (Donner et al., 2010). Since RNs and these graph-theoretical metrics are in their original application a discrete approximation of more general and continuous geometric properties of a dynamical system's underlying attractor (Zou et al., 2019), how this ultimately applies and transfers to psychometrics remains an open question as well.

Intuitively RNs can be understood as modeling the time evolution of a complex system as a trajectory through some kind of phase/state-space (Varley & Sporns, 2021)[14]. And in some form this has already been applied in psychology previously (Coco & Dale, 2014; Hasselman & Bosman, 2020; Tomashin et al., 2024; Wallot, 2019; Wallot & Leonardi, 2018; Wallot et al., 2016), where the *Cross-Recurrence Plot* (CRP) is a useful tool for analyzing the dynamics of dyads (Logrieco et al., 2026; Renjaän et al., 2026; Vanoncini et al., 2025)). *Inter-system Recurrence Networks* (IRN) have been used to study the coupling dynamics between interacting systems of physiological variables (Hasselman et al., 2023), and in Hasselman (2022) a RN is used to analyze a depression-dataset (Idiographic; N=1, T=Large) for early warning signals (Evers et al., 2024).

[13]Correlation networks are a special case (Donner et al., 2010), and Random Geometric Graphs (RGG; Duchemin and Castro, 2023) are formally equivalent (Zou et al., 2019). Also related to Multiplex Graphs (or multilayer networks; Hasselman, 2023) which can be respectively constructed from Horizontal Visibility Graphs (HVG; Lacasa et al., 2015).

[14]points to a relation between RNs and *Topological Data Analysis* (TDA; Chazal and Michel, 2021; Sizemore et al., 2019)

### Higher-order Networks

When Graph Theory is utilized to formulate our inferences problems, a large literature on higher order relations is—at least in principle—available for incorporation into future psychometric methodology as well. An introductory overview into the emerging field of networks beyond pairwise interactions like *Hypergraphs* and *Simplical-Complexes*[15] can be found in Battiston et al. (2020). Hypergraphs have already been proposed in psychometrics to overcome the limitation of only modeling pairwise interactions (Marinazzo et al., 2024), and the IM can be formulated on hypergraphs as well (Mukherjee et al., 2022; Robiglio, Di Gaetano, et al., 2025), the SBM and ER model can be too (Mukherjee et al., 2022). Related publications on hypergraphs include their application to Dynamical Systems and time-series data (Carletti et al., 2020; Delabays et al., 2025; Vaucher & Minchella, 2026) with community structure (Lotito et al., 2024; Ruggeri et al., 2024) and latent space approaches for graphs (Turnbull et al., 2023). When we attempt to model human subjective experiences—as we usually do in psychometrics—the variables of interest can originate from different levels of abstraction (e.g., items in psychometric tests referring to biological, psychological, or social variables), such a multi-level structure is also considered crucial in some theories of mind (e.g., Nested States Model (NSM; Denfield and Kyzar, 2024)) and reflected in the application of *multilayer networks* (Kivela et al., 2014) as a framework for more adequately modeling psychopathologies (de Boer et al., 2021), as well as for ecological modeling in general (Jordán, 2022), and represents yet another important methodological extension (I. Kim et al., 2020; Krishnagopal & Bianconi, 2023). In addition to all that, there is a huge literature on investigating higher-order interactions in dynamical systems and how important they are for understanding their dynamics (Battiston et al., 2021; Bick et al., 2023; Boccaletti et al., 2023; Ceria & Wang, 2023; Klein & Hoel, 2020; Lambiotte et al., 2019; Majhi et al., 2022; Malizia et al., 2024, 2025; Muolo et al., 2024; Papillon et al., 2025; Robiglio, Neri, et al., 2025; Rosas et al., 2022; Yao et al., 2021).

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[15] Have been identified as potentially appropriate for the social sciences some time ago already (Harary & Norman, 1960).

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